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A PALEOMAGNETIC POLARITY TRANSITION IN THE DEVONIAN COLUMBUS LIMESTONE OF OHIO: A POSSIBLE STRATIGRAPHIC TOOL

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ABSTRACT

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A 40-m section, including the top 1 m of the Raisin River Dolomite, 31 m of the type section of the Columbus Limestone, and 7 m of the Delaware Limestone, was sampled at 15-cm intervals for paleomagnetic stratigraphy. The Raisin River Dolomite and the first meter of the Columbus Limestone were normally magnetized (position of the paleomagnetic pole: 24°N164°E, $dp = 9.3°$, $dm = 11.3°$). The next 16 m of the Columbus Limestone shows what the author interprets to be a transition zone from normal to reversely magnetized sediments. This is followed by 22 m of reversely magnetized sediments of the Columbus Limestone (position of the paleomagnetic pole: 45°N120°E, $dp = 2.9°$, $dm = 1.6°$) and 7 m of the Delaware Limestone (position of the paleomagnetic pole: 48°N118°E, $dp = 4.0°$, $dm = 2.0°$). Since the section might not represent sufficient time to average out secular variation, these pole positions may not represent the axial geocentric dipole. The reversal should be useful as a stratigraphic marker horizon. The transition zone should be useful for a detailed study of an Early Paleozoic reversal of the earth's magnetic field. Due to the low inclination values the reversal is best seen in the declinations.

INTRODUCTION

Characteristically continental paleomagnetic studies have concentrated upon red-bed sequences and igneous rocks. This is true for the Devonian, where most of the European studies (Blackett et al., 1960; Clegg et al., 1954a,b; Creer et al., 1957; Nairn, 1960; Khramov, 1968) have been concentrated upon the Old Red Sandstone lithologies. In North America, Devonian results are scarce (Graham, 1956; Black, 1964) since over much of the country, the midwest in particular, the Devonian is represented by unaltered limestones and dolomites. Since the few available studies of limestones (Graham, 1956; McAulay and Morris, 1971; Morris, 1971; McElhinney and Opdyke, 1973) suggest that limestones may yield valuable paleomagnetic data, an

initial survey of the Middle Devonian Columbus and Delaware Limestones of central Ohio was undertaken (Martin, 1971). The results of that survey showed that the limestones and dolomites provided paleomagnetic data comparable with the data already published (Graham, 1956; Black, 1964).

The present paper presents the results of a study of the Columbus Limestone, Delaware Limestone and the Raisin River Dolomite based on 253 samples taken in the 40 m thick type section in Marblecliff Quarries, Columbus, Ohio. It is an extension of the work reported by Martin (1971) and McMahon (1974).

STRATIGRAPHY

There has been considerable debate on the correct stratigraphy of the Devonian carbonates of central Ohio. Reference here is made only to certain of the recent principal and review papers (i.e. Stauffer, 1957; Ehlers et al., 1951; Oliver, 1968; Janssens, 1968, 1969, 1970a,b; Ramsey, 1969). The succession, in Fig. 1, found in Marblecliff Quarries, is now the standard for central Ohio (Janssens, 1970a).

The Siegenian and Gedinnian, as well as the topmost Silurian, are said to be cut out by the unconformity between the Raisin River Dolomite and the Columbus Limestone. An alternative interpretation of this sequence in terms of variations in the limestone facies (Martin, in preparation) regards the unconformity as insignificant and the Raisin River Dolomite, Columbus and Delaware Limestones are considered as Devonian facies variants. These facies variants represent a change from supratidal conditions of the Raisin River Dolomite to the sub-tidal—lagoonal conditions of the Columbus and Delaware Limestones.

PALEOMAGNETIC STUDY

The carbonates in the quarry were sampled at an average interval of 15 cm over the entire 40 m thickness. In some places, because of inaccessibility, the sampling interval increased to twice that amount, but in many other places samples were spaced at as little as 5-cm intervals.

After the NRM measurements were made, a number of pilot samples were progressively demagnetized with a Schonstedt GSD-1 demagnetizer along three orthogonal axes (Fig. 2). The results of the remanence measurements, made using a Schonstedt spinner magnetometer and an x-y recorder, suggest that the component of secondary magnetization present was removed after treatment in fields of 150—250 Oe. The changes in direction as a result of cleaning are most pronounced in the magnetic inclination (Fig. 3A), however, the low paleomagnetic dips obtained after cleaning, make the reversal clearly visible only in the declinations. All cores were then cleaned at two or more stages within the interval mentioned and the cleaned results are shown in Fig. 4A.

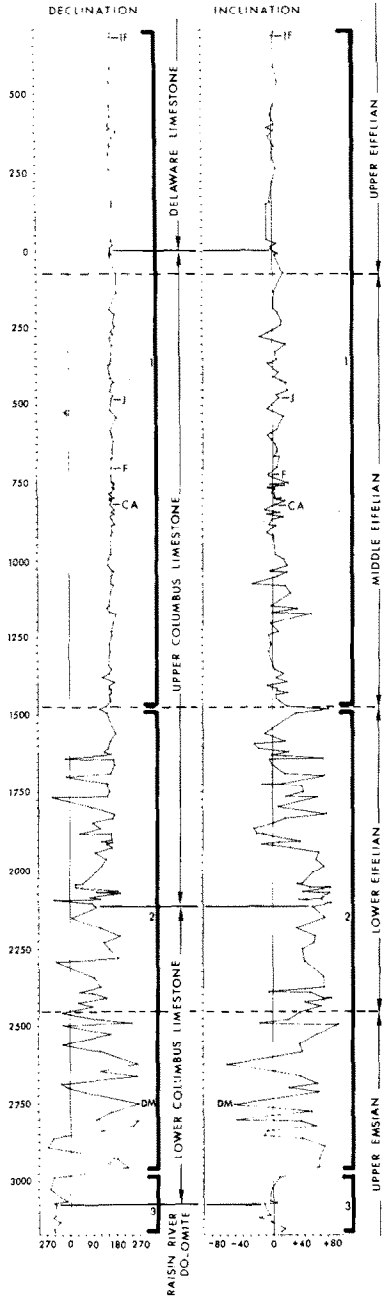


Fig. 1. Stratigraphic plot of sample declinations after 250 Oe demagnetization. The vertical scale is in centimeters using the top of the Columbus Limestone as 0 and the horizontal scale is in degrees. In the declination plot a vertical line is drawn through 0° to illustrate the fluctuations in the transition zone, and in the inclination plot the vertical line is drawn through 0° inclination. The formation boundaries and the stage boundaries (Ramsey, 1969) are shown to illustrate the location of the transition zone. The locations of samples F, J, CA, and IF from Fig. 2 and 4 are also shown. Bracket 1 indicates the samples used in the calculation of the pole position for the Columbus and Delaware Limestones, bracket 2 is the transition zone, and bracket 3 indicates the samples used for the calculation of the normal pole position (Table I).

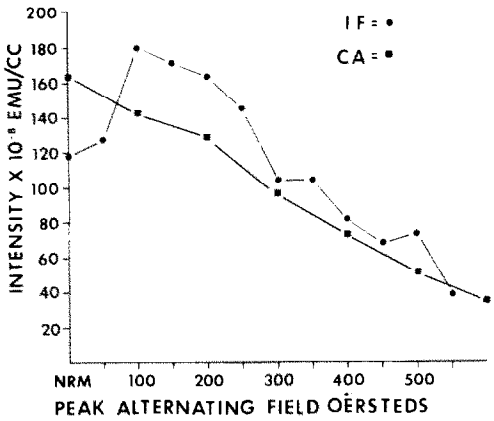


Fig. 2. AC demagnetization curves for samples IF and CA (IF = ●, CA = ■). Sample IF is from the Delaware Limestone and sample CA from the Columbus Limestone. Both are reversely magnetized.

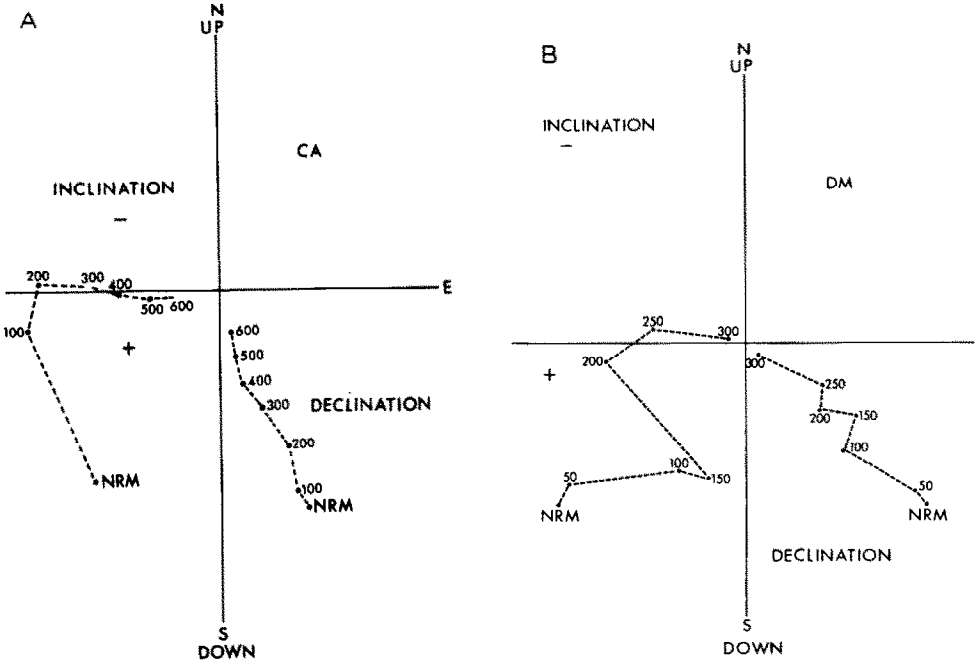


Fig. 3. Demagnetization diagram for sample CA (A) and DM (B). The plotted points in diagrams represent successive positions, in orthogonal projection, of the end of the resultant magnetization vector during progressive demagnetization. The declination plot represents a projection on the horizontal plane, while the inclination plot represents a projection on the north—south vertical plane. Numbers denote AC magnetic field intensities in oersteds.

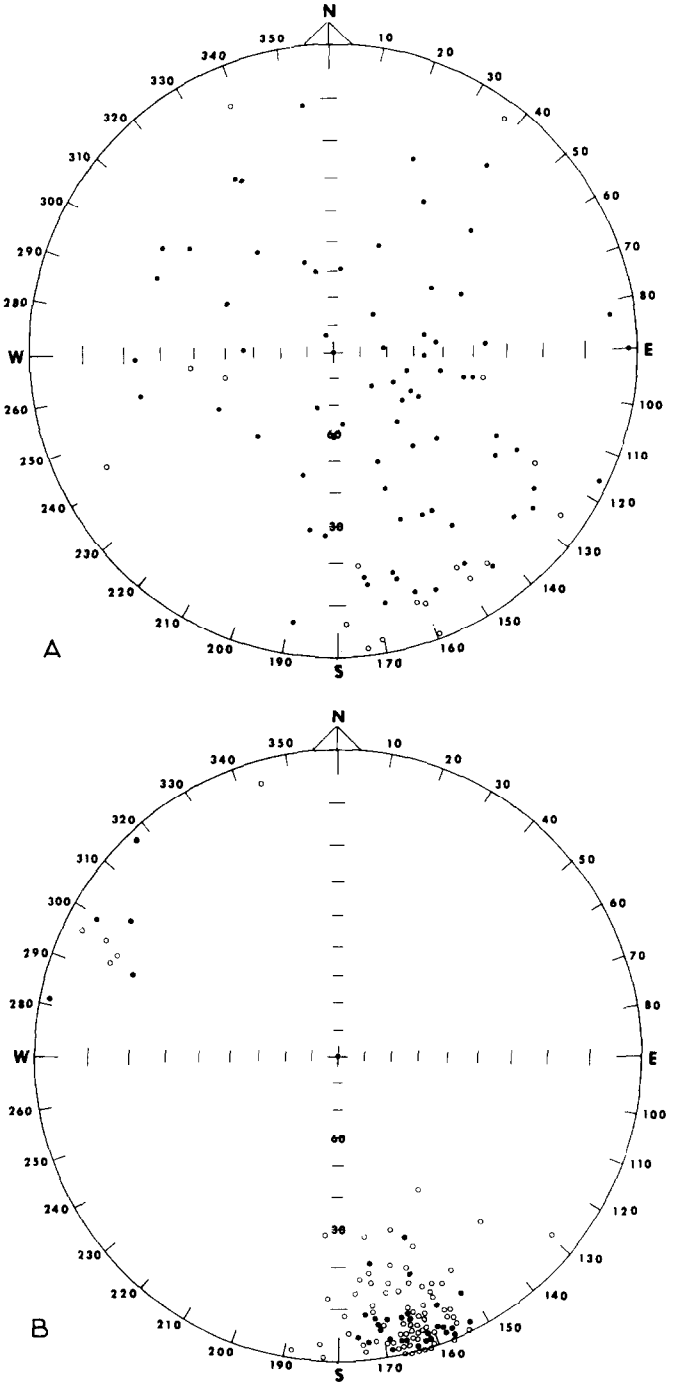


Fig. 4. Equal-area projection of cleaned results after AC demagnetization fields of 250 Oe. ● = + inclinations, ○ = — inclinations. A. Transition zone. B. All reversed and normal directions with a few overlapping points omitted from the reverse portion for clarity.

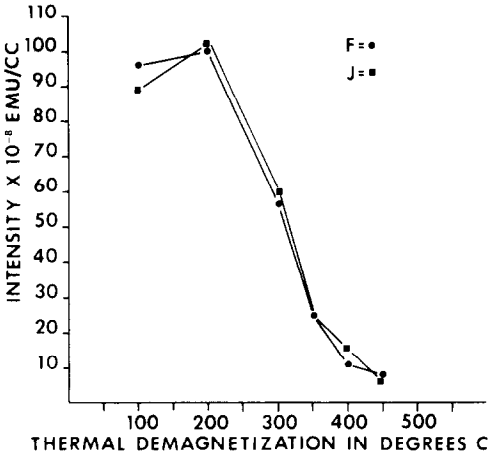


Fig. 5. Thermal-demagnetization curves for samples J and F (J = ■, F = ●) up to 450°C.

Figure 4B shows the cleaned results from the transition zone which clearly shows the large scatter and steep dips, while Fig. 3B (taken from core DM in the transition zone) demonstrates the stability of the directions after cleaning.

Identification of the carrier of the remanence is difficult optically, for in all the polished sections studied the grainsize was too small and the particles too disseminated for identification under 1350X magnification. The few discernable minerals were clearly diagenetic, they include euhedral and concretionary sulphides rimmed with hematite. Several samples were thermally demagnetized (Fig. 5), and as the figure shows the blocking temperature lies in the range 350—400° C suggesting titanomagnetites as a possible carrier of the remanence. The rapid direction changes in the transition zone (anomalous zone, Fig. 1, bracket 2) suggests the magnetization if not depositional is early post-depositional and not chemical.

Due to the large fluctuations in the transition zone, several cores were measured on two other magnetometers, giving similar results. Although no absolute intensity measurements were made, the relative intensity of samples in the normal, transition, and reversed zones were approximately the same. Only the Delaware Limestone as a whole was stronger than the other samples. All samples were well within the sensitivity of the magnetometers used, and therefore the transition-zone fluctuations are not thought to be machine-generated.

DISCUSSION OF RESULTS

There are several results from this study. The first is a confirmation of the value of limestones in paleomagnetic work. The second is the demonstration

of the existence of both normal and reversed magnetizations in the section, although it is unfortunate that so little of the limestone and dolomite in which normal magnetization is found is exposed in the quarry. Third, the normal polarity found in this quarry represents the only normal polarity found in the Devonian of North America, although the Russians (Khramov, 1968) have reported them in their studies. The reversed magnetization in the greater part of the Columbus Limestone and the overlying Delaware Limestone is remarkably constant (Table I). The mean direction of magnetization for each limestone (the transition zone (anomalous zone) excluded) is listed in Table I. Whether the time duration of each limestone was sufficient to average out secular variation is questionable in view of the uncertain rate of carbonate deposition (estimates range from 17,000 years to 425,000 years for the 22 m, based upon published depositional rates) (Cloud, 1959; Newell, 1955).

The fourth is the discovery of a transition zone 16 m thick between the normal and reversed zones. Its base lies 1.5 m above the base of the Columbus Limestone in Marblecliff Quarries. Transition zones may be of value as a record of the behavior of the geomagnetic field during a polarity reversal. Relatively few transition zones are recorded in the literature (York et al., 1971; Goldstein et al., 1969; Larson et al., 1971), the best being the deliberate search for them in Cenozoic intrusions carried out by Dodson et al. (1973). The most detailed study to date is that of Helsley and Steiner (1973) of an anomalous zone in Triassic sandstone. The advantage offered by a transition zone in some limestones as distinct from other sedimentary sequences or lava-successions is the continuity of record.

The lagoonal conditions under which the limestone formed (Martin, in preparation), without any large influx of sediment or marked changes of sedimentation rate in so far as can be determined, provide the best changes of viewing continuous sedimentation. It is notoriously difficult to estimate carbonate sedimentation rates with accuracy and dangerous to apply modern estimates to ancient conditions; however, they may be used to provide an order of magnitude estimate of the elapsed time represented by 16 m of limestone. Cloud (1959) estimated that as much as 100 cm/1000 years of shallow-water carbonate could form, while Newell (1955) gave 4 cm/1000 years. The more rapid rate was applied to recent carbonate deposition. These figures suggest approximate limits to the duration of the transition zone of 342,000 years and 13,716 years, respectively. Although there are no absolute measurements of the duration of a transition zone, the estimates usually quoted are considerably less which may imply that carbonate sedimentation rate during the Devonian was close to the higher rate.

Assuming a uniform sedimentation rate, Fig. 1 illustrates the erratic behavior of the field during transition from normal to reversed polarity. The characteristically low inclination of the ordinary field (both normal and reversed) being replaced by high inclination values and erratic declinations with occasional and rapid but brief polarity switches. There is a suggestion

TABLE I

Summary of statistical data

	Magnetic direction						Pole position			
	Decl.	Incl.	N.	Alpha 95	K.	R.	Lat. N	Long. E	Delta P	Delta M
All samples Columbus, Delaware, (exclusive of transition zone)	164.2°	4°	119	2.0°	40.7	114.15	45.5°	119.8°	2.5°	1.4°
Columbus Limestone (exclusive of transition zone)	163.8°	4.9°	98	2.3°	37.1	95.39	45.0°	120.1°	2.9°	1.6°
Delaware Limestone	166.1°	-0.3°	21	3.0°	110.8	20.82	48.2°	118.0°	4.0°	2.0°
All normal samples Columbus Limestone and Raisin River Dolomite	302.4°	0.6°	10	12.4°	16.0	9.44	24.0°	164.5°	9.3°	11.3°

that the duration and frequency of these polarity excursions diminishes towards the top of the transition zone.

The transition zone provides a test of the two stratigraphic ideas advanced earlier. For, if the facies hypothesis of Martin (in preparation) is valid, then it is unlikely that the transition zone will be in the same stratigraphic position in the Columbus Limestone facies elsewhere, while the reverse will be anticipated if the standard interpretation of the unconformity is correct. This is currently being pursued.

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